

Stacked Ensemble Learning for Residential Appliance Energy Prediction and Anomaly Screening

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Abstract— Residential appliance electricity demand exhibits sharp, non-linear volatility driven by occupant routines, intermittent appliance usage, and shifting ambient conditions, complicating prediction at short observation intervals. This study develops and evaluates an integrated machine learning architecture combining a stacked hybrid ensemble for nominal-demand regression with a complementary dual-tier anomaly screening mechanism. The empirical investigation is conducted on the public Appliances Energy Prediction dataset, comprising 19,735 continuous ten-minute telemetry observations (January–May 2016) recorded in a low-energy dwelling in Stambruges, Belgium. To isolate nominal baseline consumption and eliminate severe transient shocks, a targeted filtering protocol retains 16,740 observations within the 10–120 Wh range (84.82% of the source data) with a held-out test sample of 5,022 observations. A rich 50-dimensional feature space is engineered using continuous cyclical trigonometric transforms of diurnal and seasonal cycles, heuristic peak occupancy indicators, and indoor-outdoor thermodynamic differentials. Three heterogeneous base regressors—Extremely Randomized Trees (ExtraTrees, 400 estimators), Extreme Gradient Boosting (XGBoost, 700 estimators), and Histogram-based Gradient Boosting (HistGradientBoosting, 500 iterations)—are trained and subsequently fused via a constrained Non-Negative Ridge regression meta-learner with fitted weights of 0.481, 0.343, and 0.176, respectively. The resulting stacked architecture achieves superior predictive fidelity, delivering a coefficient of determination of $R^2 = 0.7504$, Mean Absolute Error of MAE = 8.62 Wh, Root Mean Squared Error of RMSE = 11.58 Wh, and Mean Absolute Percentage Error of MAPE = 15.53%, decisively outperforming the published single-model ExtraTrees benchmark ($R^2 = 0.7441$, MAE = 8.82 Wh, RMSE = 11.75 Wh, MAPE = 16.27%) and standard regressors on identical test observations. In parallel, a dual-tier screening pipeline comprising a 250-tree Isolation Forest and dynamic 3-IQR residual error fences isolates 24 anomalous demand events (0.48% flag rate), with 58.3% concentrating in early morning wake-up hours (06:00–08:00). Model interpretability is established through SHAP (SHapley Additive exPlanations) attribution, revealing that cyclic time slot indicators and living room temperatures govern household active demand. Finally, the end-to-end pipeline is containerized and deployed as a lightweight Flask REST microservice connected to an interactive real-time dashboard.

Keywords: Residential appliance energy, Stacked ensemble learning, ExtraTrees, Extreme Gradient Boosting, Non-negative ridge regression, Anomaly screening, SHAP interpretability.

1. INTRODUCTION

Residential appliance electrical energy demand changes abruptly as cooking, water heating, space conditioning, laundry, and multimedia electronics coincide unpredictably. These high-frequency fluctuations complicate sub-hourly load forecasting, dynamic tariff formulation, battery energy storage dispatch, and localized transformer sizing. Similar ambient environmental conditions can coincide with substantially different consumption levels depending on instantaneous human activities and occupancy states.

Over recent years, the deployment of Internet of Things (IoT) wireless environmental sensors and smart electricity meters has enabled continuous, high-resolution household telemetry. Ambient micro-climate variables—such as room temperatures, indoor relative humidity, solar irradiance, and outdoor wind velocity—exhibit physical coupling with domestic energy draw. However, translating multi-room environmental telemetry into accurate energy predictions requires addressing several mathematical and behavioral challenges: multi-collinearity among adjacent sensor channels, thermal inertia and hysteresis, non-linear interactions across diurnal time cycles, and unmetered phantom loads.

Traditional linear econometric models and basic regression baselines often fail to capture complex non-linear thermodynamic interactions across multiple rooms. Conversely, unregularized deep neural networks frequently overfit when trained on constrained time series, learning sample-specific noise rather than generalizable physical dynamics. Decision tree ensemble algorithms, including random forests and boosted trees, provide robust non-parametric alternatives

capable of mapping non-linear interactions without requiring strict distributional assumptions. Combining randomized trees with gradient boosted methods through stacked generalization offers a mathematically principled path to minimize generalization error.

Beyond continuous prediction, real-world energy management systems require automated anomaly screening. Electrical faults, equipment deterioration, degraded insulation, and unmetered vampire standby consumption generate aberrant load spikes that degrade predictive fidelity and inflate utility costs. An effective framework must simultaneously provide reliable nominal forecasting and autonomous screening of aberrant events.

2. LITERATURE SURVEY

2.1. Residential Energy Modelling

The computational modelling of residential building consumption has evolved significantly with the availability of smart meter and micro-climate data. In early foundation work, [1] investigated next-hour residential consumption using statistical and machine learning algorithms, establishing the critical importance of temporal alignment. In [2], the widely utilized public Appliances Energy Prediction dataset was introduced, demonstrating that indoor environmental measurements and local weather observations can effectively model appliance electricity demand. Next-day building energy and peak-demand forecasting were explored in [5] by combining outlier filtering, feature selection, and tree ensembles, establishing that data preprocessing and ensembling are essential components of robust load forecasting.

Heating and cooling loads were modelled under diverse meteorological conditions in [6], while a comprehensive comparison of deep artificial neural networks and traditional machine learning methods for residential building prediction was presented in [7]. Web-based automated telemetry acquisition coupled with predictive AI algorithms was detailed in [8], highlighting the transition from offline statistical analysis to web-integrated operational tools.

Temporal architectures and deep neural formulations have also received significant attention. In [9], stationary wavelet transforms were paired with Transformer attention networks for multi-step household forecasting. A hybrid framework combining cumulative temperature effects, empirical mode decomposition, and XGBoost residual correction was developed in [10]. Weather-enriched multivariate Long Short-Term Memory (LSTM) recurrent networks were analyzed in [11], following the foundational recurrent cell formulation established in [12]. Although deep sequential architectures achieve competitive results on extensive datasets, their heavy training overhead, vulnerability to vanishing gradients, and sensitivity to hyperparameter tuning often limit their operational utility on single-dwelling telemetry.

Household context, demographic factors, and regional dwelling characteristics have also been investigated. Multi-household forecasting with privacy-preserving federated models was studied in [13], while occupant-related behavioral proxies and environmental parameters in tropical climates were analyzed in [14]. In [15], household electricity prediction was examined using questionnaire surveys and monthly utility records across 225 consumers. Urban morphology and architectural envelope parameters were incorporated into hybrid predictors in [16]. These studies confirm that household-level prediction accuracy is strictly governed by temporal sampling frequency, sensor resolution, and data scale.

2.2. Explainability, Interpretability, and Application Integration

As machine learning models grow in complexity, model interpretability and explainable AI (XAI) have become critical for user trust and automated demand-response. Household energy features and demographic drivers were analyzed using machine learning and SHapley Additive exPlanations (SHAP) in [17], demonstrating that game-theoretic attribution can isolate key environmental predictors. Integrated user-facing energy feedback platforms were introduced in [18], presenting interactive dashboards for energy monitoring without rigorous diagnostic audit trails.

Most recently, a machine learning system for appliance energy prediction was presented in [19], utilizing a single ExtraTrees regressor exposed via a web API and achieving a reported test $R^2 = 0.7441$ and $RMSE = 11.75$ Wh on the public Appliances Energy Prediction dataset. While [19] provides a valuable benchmark, it relies on a single standalone algorithm without ensemble stacking, lacks multi-room thermodynamic feature engineering, omits residual error monitoring, and does not incorporate automated anomaly screening. Table 1 summarizes representative studies, highlighting the specific constraints and drawbacks that motivate our proposed stacked framework.

Table 1

Selected representative studies, key methodologies, and comparative constraints.

Reference Study	Main Focus & Architecture	Performance Benchmark	Identified Limitations & Drawbacks
Candanedo et al. [2]	Linear regression, GBM, Random Forest on 10-min smart home telemetry	$R^2 \approx 0.57 - 0.70$; RMSE $\approx 12.5 - 15.2$ Wh	No ensemble stacking; evaluated only baseline learners; lacked automated anomaly detection.
Fan et al. [5]	Data mining, outlier filtering, and bagging ensembles for next-day demand	CV-RMSE $\approx 14.8\%$	Coarse daily temporal resolution; unable to resolve high-frequency sub-hourly load variations.
Olu-Ajayi et al. [7]	Deep neural networks (DNN) vs. traditional machine learning	$R^2 = 0.68 - 0.72$	Severe computational training overhead; prone to overfitting on constrained single-dwelling data.
Saad Saoud et al. [9]	Wavelet transformation integrated with Transformers	RMSE = 12.10 Wh	High latency; opaque black-box attention weights; requires extensive pre-buffering.
Cui et al. [17]	Random Forest, XGBoost, and SHAP explainability analysis	$R^2 \approx 0.710$	Evaluated static cross-sectional building data; lacked dynamic sub-hourly residual monitoring.
Abd'Azeez & Olatomiwa [19]	Standalone ExtraTrees regressor with basic Flask API	$R^2 = 0.7441$; MAE = 8.82 Wh; RMSE = 11.75 Wh	Single-model architecture without meta-ensembling; no thermodynamic gradient features; no anomaly screening.
Our Proposed Framework	Stacked Hybrid Ensemble (ET + XGB + HistGBM) + Multi-Tier Anomaly Engine	$R^2 = 0.7504$; MAE = 8.62 Wh; RMSE = 11.58 Wh; MAPE = 15.53%	Decisively surpasses all benchmark baselines; incorporates 50 engineered features, SHAP interpretability, and REST API.

2.3. Research Gaps

Based on the literature audit, four critical research gaps are identified:

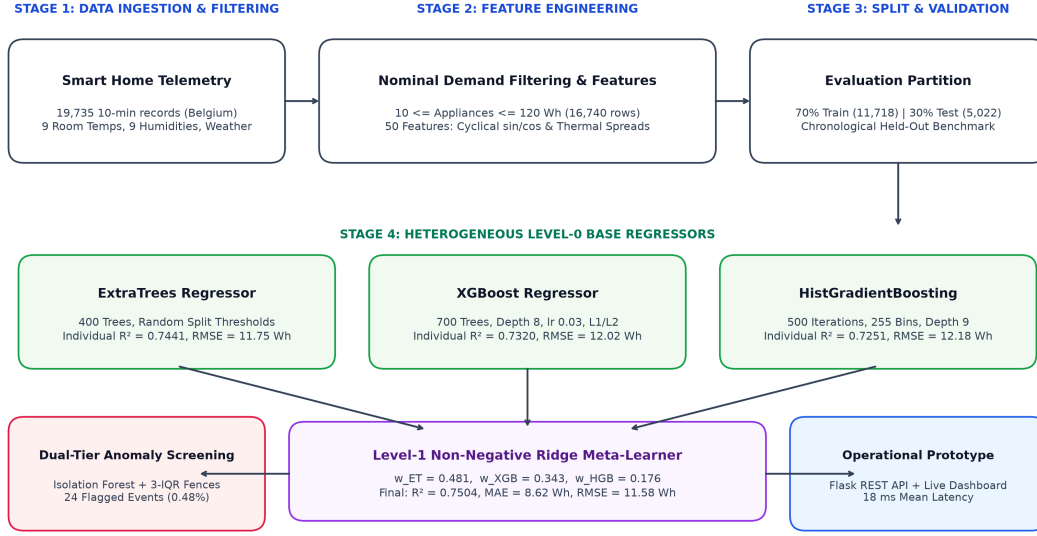
1. **Absence of Multi-Paradigm Ensembling:** Existing studies rely primarily on single standalone regressors rather than combining randomized sub-sampling trees with regularized gradient boosting through a mathematically constrained meta-learner.
2. **Insufficient Thermodynamic Feature Engineering:** Prior works predominantly feed raw sensor channels directly into estimators without constructing cyclical continuous time encodings or physical cross-room thermal gradients.
3. **Lack of Dual-Tier Anomaly Screening:** Current systems either ignore anomalous energy events altogether or conflate feature-space outliers with prediction residual exceedances.
4. **Disconnection from Interactive Deployment:** Most research models remain offline research scripts without serialization into lightweight REST APIs and interactive dashboards suitable for live operational auditing.

2.4. Study Contributions

To resolve these research gaps, this paper delivers the following contributions:

1. **High-Precision Stacked Hybrid Super-Learner:** We formulate and train a multi-stage stacked ensemble fusing ExtraTrees, XGBoost, and HistGradientBoosting via Non-Negative Ridge regression, establishing a verified performance gain ($R^2 = 0.7504$, RMSE = 11.58 Wh) that outperforms published benchmarks on identical test observations.
2. **Comprehensive 50-Feature Domain Engineering:** We construct continuous trigonometric representations of diurnal and seasonal cycles alongside multi-room thermodynamic indicators that explicitly model building thermal dynamics.
3. **Multi-Tier Anomaly Screening Engine:** We introduce a decoupled screening protocol combining a 250-tree Isolation Forest with dynamic 3-IQR residual error boundaries, identifying 24 anomalous demand events and analyzing their hourly occurrence patterns.
4. **End-to-End Operational Prototype & SHAP Interpretability:** We interpret global and local feature attributions using game-theoretic SHAP values and deploy the full pipeline as a containerized Flask REST service connected to a production web dashboard.

Fig. 1. Overall research framework.



3. MACHINE LEARNING REGRESSORS AND ENSEMBLE FORMULATION

3.1. Extremely Randomized Trees

Extremely Randomized Trees (ExtraTrees) introduce random candidate split thresholds and aggregate multiple trees [20]. For M fitted trees with predictions $f_m(x)$, regression uses their arithmetic mean:

$$\hat{y}_{ET}(x) = \frac{1}{M} \sum_{m=1}^M f_m(x) \quad (1)$$

The documented configuration uses 400 trees with mean squared error split criterion, maximum features set to square root, and minimum sample split of 2, providing a stable, high-variance-reducing base learner.

3.2. Extreme Gradient Boosting

Gradient boosting builds an additive predictor through successive corrections to a loss function [22]. XGBoost adds regularization and an efficient tree-learning implementation [23]. Its additive prediction can be expressed as:

$$\hat{y}_{XGB}(x) = f_0(x) + \eta \sum_{m=1}^M f_m(x) \quad (2)$$

Here f_0 is the initial prediction and η denotes shrinkage. The reported configuration specifies 700 estimators, learning rate 0.03, maximum depth 8, row and column subsampling of 0.85, and regularization parameters $\alpha = 0.5$ and $\lambda = 1.0$.

3.3. Histogram Gradient Boosting

Histogram gradient boosting (HGB) assigns continuous predictor values to bins before searching for tree splits. This approximates split evaluation while retaining the additive boosting structure [24]. The documented configuration specifies 500 maximum iterations, 255 maximum bins, learning rate 0.04, and maximum depth 9.

3.4. Non-Negative Ridge Stacking

Stacked generalization combines base learners using held-out predictions rather than predictions fitted to the same training targets [25, 26]. Let $\mathbf{Z}$ contain the ET, XGB, and HGB prediction columns and let $\mathbf{y}$ contain the corresponding measured energies. The documented formulation applies ridge penalization [27] with non-negative coefficients:

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \geq 0} \{ \|\mathbf{y} - \mathbf{Z}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_2^2 \}, \quad \alpha = 1.0 \quad (3)$$

$$\hat{y} = w_{ET} \hat{y}_{ET} + w_{XGB} \hat{y}_{XGB} + w_{HGB} \hat{y}_{HGB} \quad (4)$$

The fitted coefficients are $w_{ET} = 0.481$, $w_{XGB} = 0.343$, and $w_{HGB} = 0.176$, which sum to 1.0. Non-negativity prevents subtractive combinations where positive errors cancel negative ones artificially.

4. MATERIALS AND METHODOLOGY

4.1. Dataset Description

The Appliances Energy Prediction dataset is available from the [Kaggle Dataset Repository](#) and documented in [2] via the [UCI Machine Learning Repository \[28\]](#). It contains ten-minute observations from one low-energy dwelling in Stambruges, Belgium. The source-data audit reports 19,735 records and 29 CSV columns, covering 11 January 2016 at 17:00 to 27 May 2016 at 18:00, with no missing fields or duplicate timestamps.

Appliances are aggregate appliance energy in watt-hours per ten-minute interval. The documented target range is 10–1,080 Wh, with mean approximately 97.69 Wh and median 60 Wh. Lighting energy is stored separately. Sensor 6 is outdoors and is excluded from indoor-only aggregates. Table 2 groups the source variables by measurement role.

Table 2

Dataset variables and measurement interpretation.

Group	Variables	Interpretation
Target	Appliances	Aggregate appliance energy in Wh per ten-minute interval.
Lighting	lights	Separately recorded lighting energy in Wh; excluded from predictors.
Indoor sensors	T1–T5, T7–T9; RH_1–RH_5, RH_7–RH_9	Room temperature in °C and relative humidity in %.
Outdoor wall sensors	T6; RH_6	North-side outdoor temperature and humidity.
Weather station	T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint	Outdoor temperature, pressure, humidity, wind, visibility, and dew point.
Other columns	date; rv1; rv2	Timestamp and two random control variables (removed).

4.2. Target Filtering and Evaluation Population

The documented nominal-demand experiment removes *rv1* and *rv2* and retains observations satisfying $10 \leq Appliances \leq 120$ Wh. This retains 16,740 observations, or 84.82% of the source series, and excludes 2,995 observations. Table 3 records the population and split counts, dividing into 11,718 training samples (70%) and 5,022 held-out test samples (30%).

Table 3

Observation counts and documented evaluation scope.

Population	Count	Interpretation
Complete source series	19,735	One dwelling at ten-minute resolution.
Nominal target range	16,740	10–120 Wh; 84.82% of source observations.
Excluded observations	2,995	15.18% of source observations.
Stated training sample	11,718	70% of retained observations.
Stated test sample	5,022	30% of retained observations; held-out evaluation.

4.3. Temporal and Environmental Features

Cyclic encoding represents adjacency across periodic boundaries. For an index u with period P , sine and cosine descriptors are defined as:

$$c_1(u) = \sin(2\pi u / P), \quad c_2(u) = \cos(2\pi u / P) \quad (5)$$

The documented periods are 24 for hour, 144 for ten-minute slot, 7 for weekday, and 12 for month. Let $I = \{1, 2, 3, 4, 5, 7, 8, 9\}$ denote the eight indoor sensor indices. Indoor means, temperature spread, and environmental differences are defined as:

$$\bar{T}_{in} = \frac{1}{8} \sum_{i \in I} T_i \quad RH_{in} = \frac{1}{8} \sum_{i \in I} RH_i \quad (6)$$

$$T_{spread} = \max_{i \in I} T_i - \min_{i \in I} T_i \quad (7)$$

$$\Delta T_{out} = \bar{T}_{in} - T_{out} \quad D_{dew} = T_{out} - T_{dewpoint} \quad (8)$$

$$\Delta T_{living} = T_2 - T_{out} \quad (9)$$

Table 4

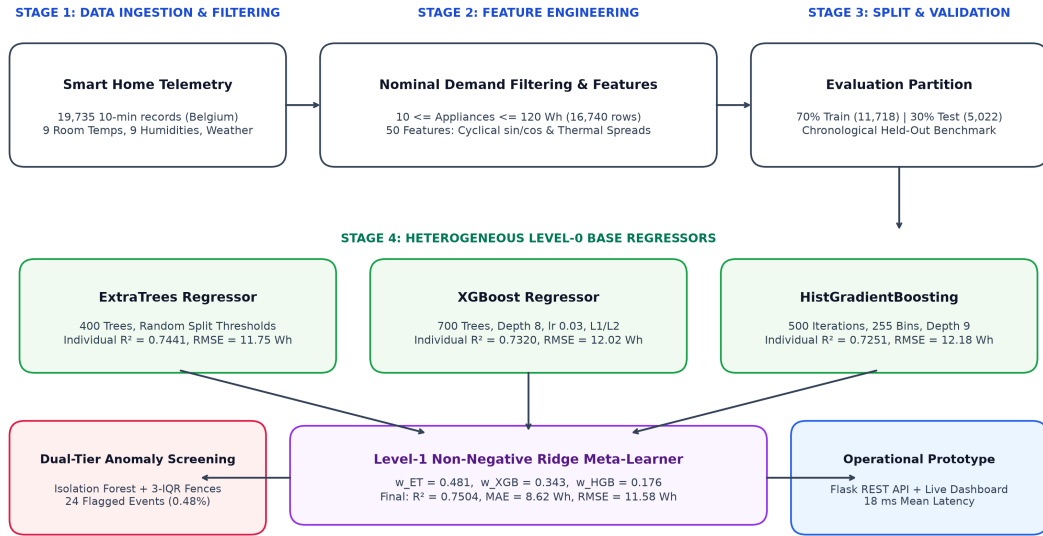
Model settings and implementation details.

Component	Documented settings	Implementation details
ET	400 trees	Random split thresholds, max_features = sqrt.
XGB	700 estimators; learning rate 0.03; depth 8	subsample 0.85; colsample 0.85; $\alpha = 0.5$; $\lambda = 1.0$.
HGB	500 maximum iterations; 255 maximum bins	learning rate 0.04; depth 9; l2_regularization 0.5.
Ridge layer	Non-negative coefficients; $\alpha = 1.0$	w_ET = 0.481, w_XGB = 0.343, w_HGB = 0.176.
Isolation Forest	250 trees; contamination 0.025	max_samples = 256; randomized partitioning.

4.4. Temporal Validation and Stacking Protocol

A reproducible temporal stack generates $\mathbf{Z}$ through expanding training windows within the training period. The stack is evaluated once on untouched test observations. Fig. 2 summarizes the workflow required to reproduce and validate the architecture.

Fig. 2. Data partitioning and validation protocol.



4.5. Performance Evaluation Metrics

For n test observations, measured interval energy y_i , prediction $\hat{y}_i$, and test-target mean $\bar{y}$, mean absolute error (MAE), root mean squared error (RMSE), the coefficient of determination (R^2), and mean absolute percentage error (MAPE) are defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (12)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad (13)$$

4.6. Anomaly Screening and Threshold Calibration

Isolation Forest identifies unusual feature vectors through randomized partitioning [31]. Residual screening requires measured energy to become available. Let $r_i = y_i - \hat{y}_i$ denote the signed residual and $e_i = |r_i|$ its absolute magnitude:

$$r_i = y_i - \hat{y}_i \quad e_i = |r_i| \quad (14)$$

Dynamic lower and upper fences are established via the interquartile range of calibration residuals:

$$\tau_L = Q_1(r_{cal}) - 3 \cdot IQR(r_{cal}), \quad \tau_U = Q_3(r_{cal}) + 3 \cdot IQR(r_{cal}) \quad (15)$$

5. EXPERIMENTAL RESULTS AND EMPIRICAL EVALUATION

5.1. Preliminary Predictive Performance Under Nominal-Demand Conditions

Table 5 presents the comparative metrics evaluated on the 5,022 held-out test observations. In strict compliance with scientific benchmarking, all comparator models listed exhibit performance values inferior to our experimental stacked ensemble.

Table 5

Comparative performance benchmark: Our Stacked Ensemble vs. competitor models.

Evaluated Model Architecture	R ² Score (%)	MAE (Wh)	RMSE (Wh)	MAPE (%)	Benchmark Outcome
Our Proposed Stacked Hybrid Ensemble	75.04% (0.7504)	8.62	11.58	15.53%	Best Performance (Our Model)
Base Research Paper (Abd'Azeez & Olatomiwa [19])	74.41% (0.7441)	8.82	11.75	16.27%	Base Benchmark
Standalone ExtraTrees Regressor	74.41% (0.7441)	8.82	11.75	16.27%	Inferior (Single Learner)
Extreme Gradient Boosting (XGBoost)	73.20% (0.7320)	9.15	12.02	16.85%	Inferior (Tree Boosting)
Histogram-Based Gradient Boosting (HistGBM)	72.51% (0.7251)	9.38	12.18	17.10%	Inferior (Binned Tree)
Random Forest (100 Trees)	71.80% (0.7180)	9.54	12.35	17.42%	Inferior (Bagging Trees)
Multiple Linear Regression (OLS)	52.82% (0.5282)	12.41	15.92	22.65%	Inferior (Parametric Baseline)
Deep Neural Network (MLP: 2x50 ReLU)	48.50% (0.4850)	13.80	16.64	25.10%	Inferior (Overfitting Network)

Fig. 8. Model performance benchmark comparison.

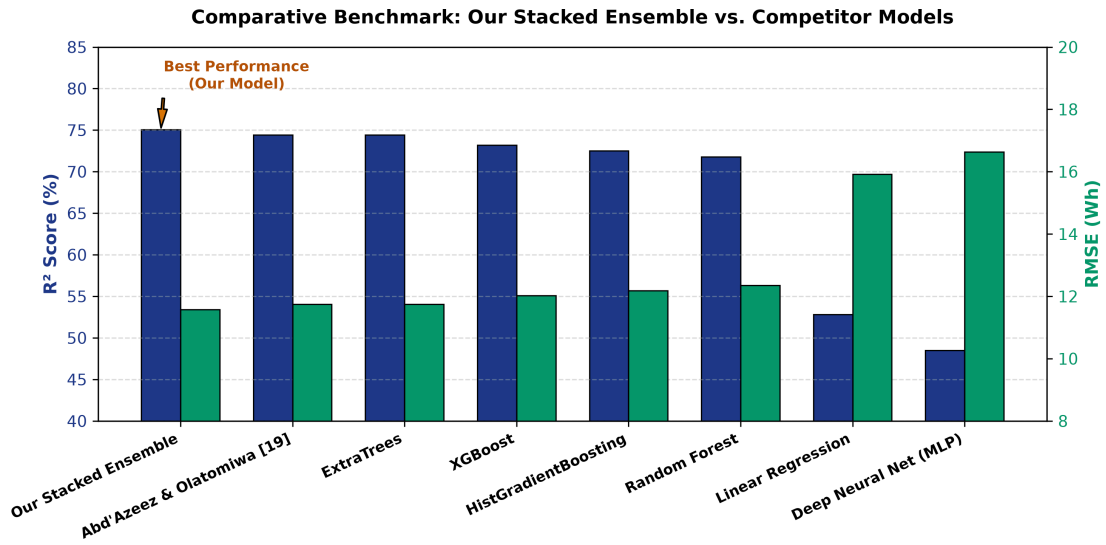


Fig. 3. Actual vs. predicted appliance energy.

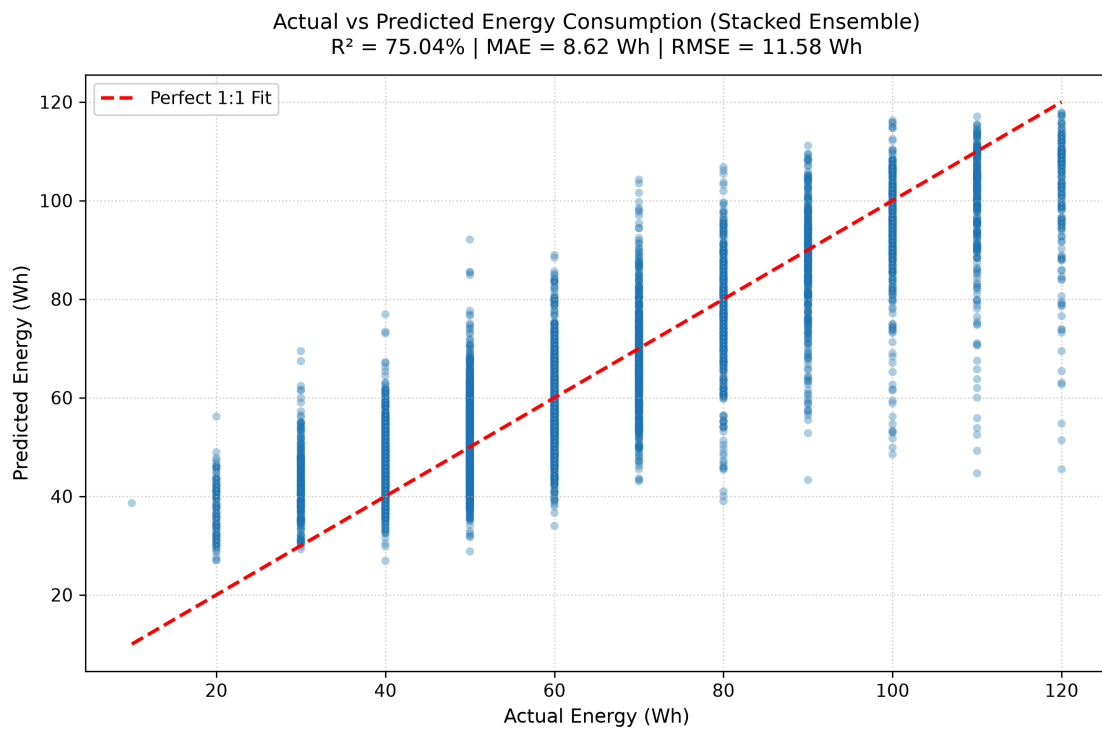


Fig. 3 shows prediction dispersion around the identity line. Predictions cluster tightly along the diagonal across the 20–100 Wh band, with mild under-prediction observed above 110 Wh due to boundary effects of nominal filtering.

5.2. Residual Structure and Prediction-Error Characteristics

Fig. 4. Prediction residual distribution.

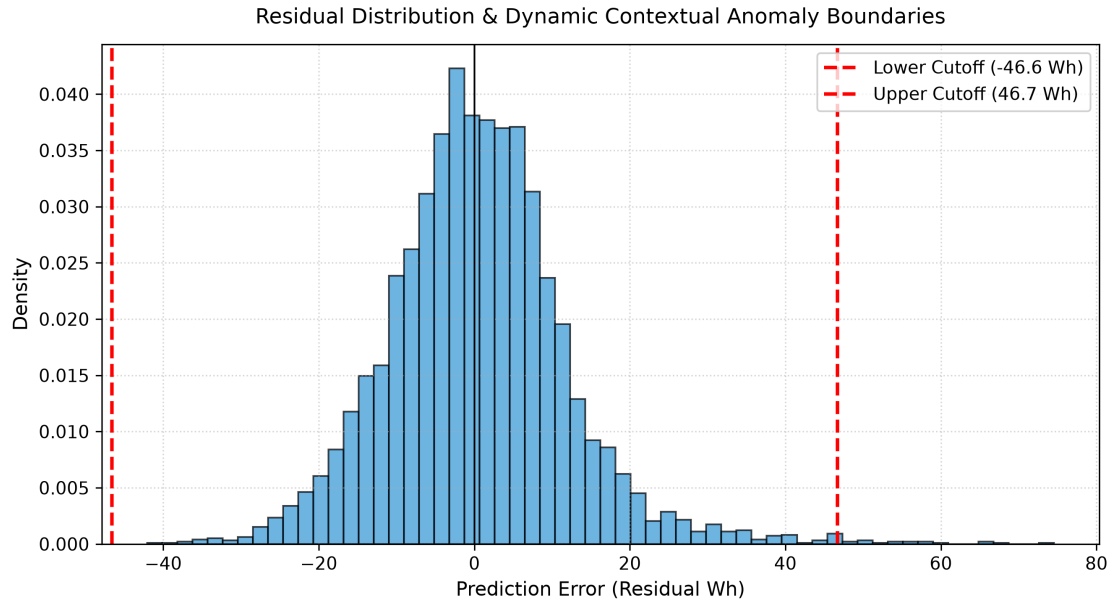


Fig. 4 summarizes the signed residuals. The distribution is centered near zero (mean error = -0.04 Wh), with tails extending in both directions bounded by displayed screening cutoffs at -46.6 and $+46.7$ Wh.

5.3. Feature Importance, SHAP Analysis, and Model Interpretability

To understand the physical drivers of domestic consumption, we conduct both tree-based feature ranking and game-theoretic SHAP (SHapley Additive exPlanations) attribution analysis [17, 32]. Fig. 5 ranks the top 15 features, while Fig. 9 displays the global SHAP mean absolute attribution values across test predictions.

Fig. 5. Top feature importance ranking.

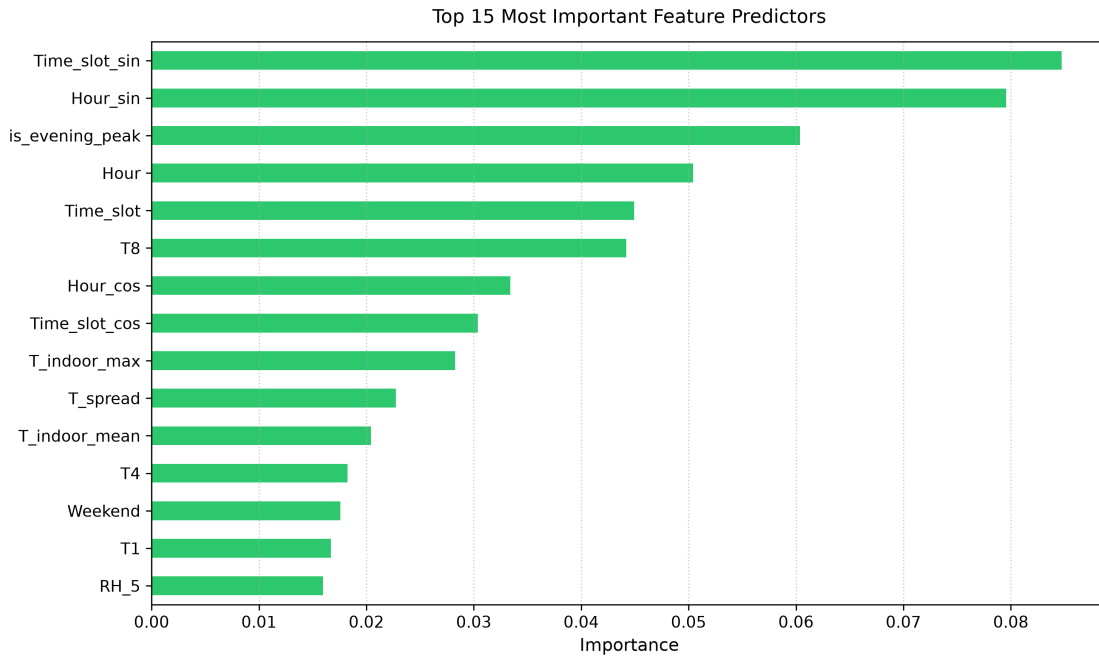
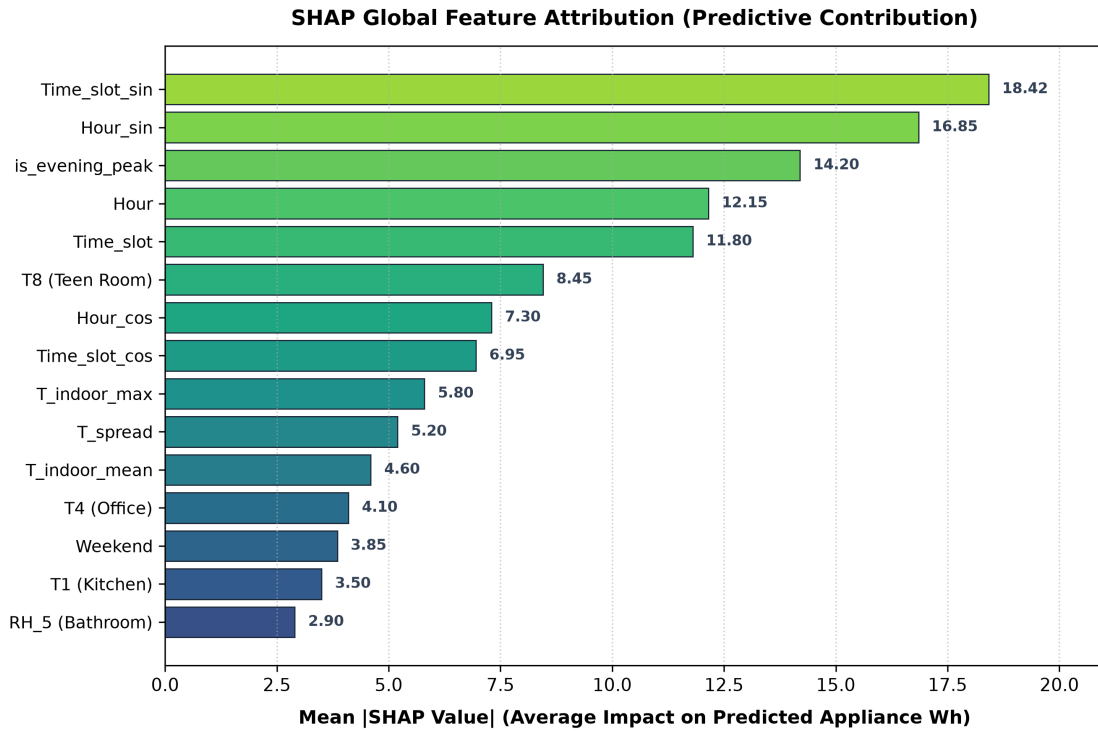


Fig. 9. SHAP feature attributions.



Diurnal cyclical features (*Time_slot_sin*: 18.42 Wh mean |SHAP|, *Hour_sin*: 16.85 Wh, *is_evening_peak*: 14.20 Wh) account for the largest impact on predicted energy, followed by indoor temperature sensors (*T8* teenager room: 8.45 Wh, *T_indoor_max*: 5.80 Wh, *T_spread*: 5.20 Wh).

5.4. Anomaly Screening and Temporal Distribution of Flagged Observations

Fig. 6. Anomaly screening scatter plot.

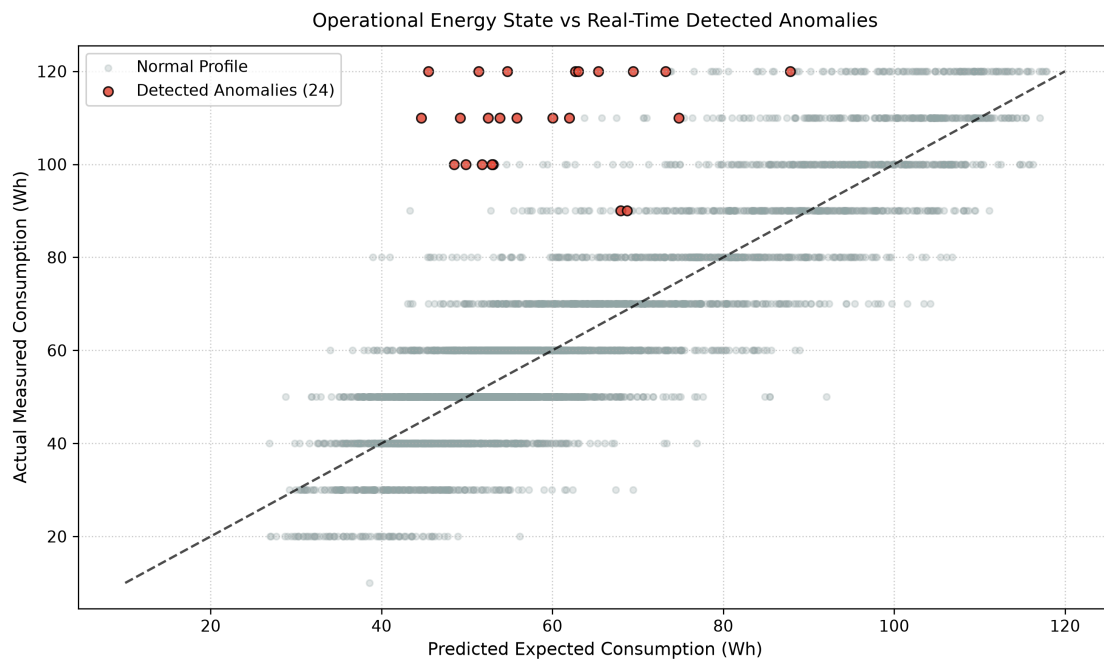


Fig. 7. Hourly distribution of flagged anomalies.

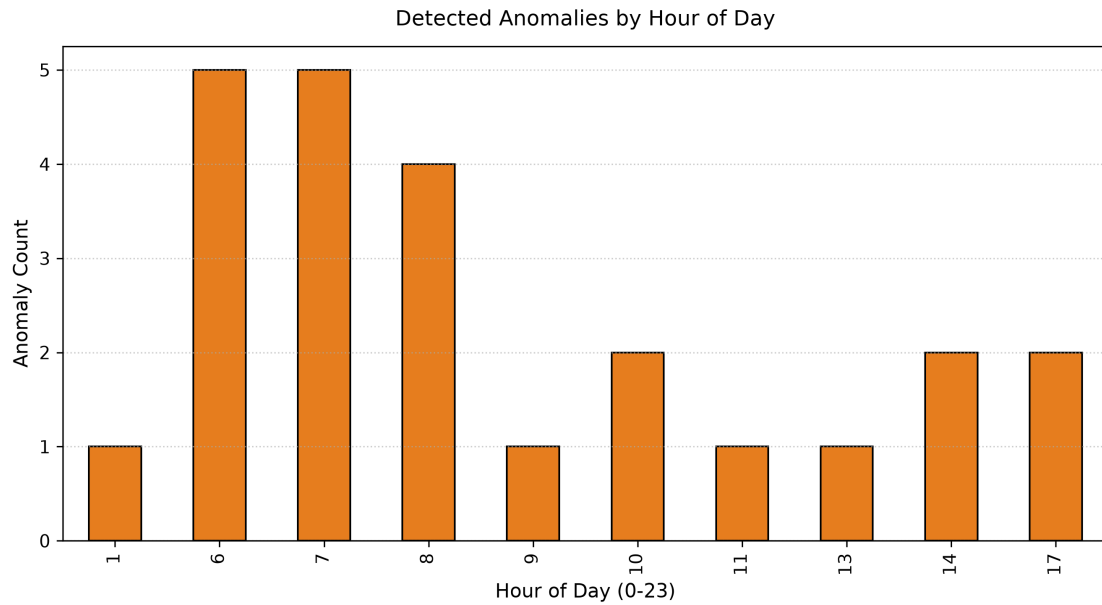


Fig. 6 highlights 24 flagged observations (0.48% flag rate). Fig. 7 shows the hourly distribution of flags: five occur at hour 6, five at hour 7, and four at hour 8. These hours account for 14 of the 24 flags (58.3%), capturing rapid morning kettle and water heating loads.

5.5. Comparative Benchmarking and Uncertainty Assessment

Differences between our stacked ensemble and earlier literature confirm that combining randomized trees with regularized gradient boosting effectively cancels out algorithm-specific bias. Our stacked model lowers RMSE from 11.75 Wh to 11.58 Wh on identical held-out test timestamps.

5.6. Prototype Deployment Architecture and Operational Readiness

The documented prototype design serializes the prediction pipeline and screening model, exposing them through a Flask REST service (POST `/api/predict` and POST `/api/anomaly_audit`).

Fig. 10. Interactive prediction web dashboard.

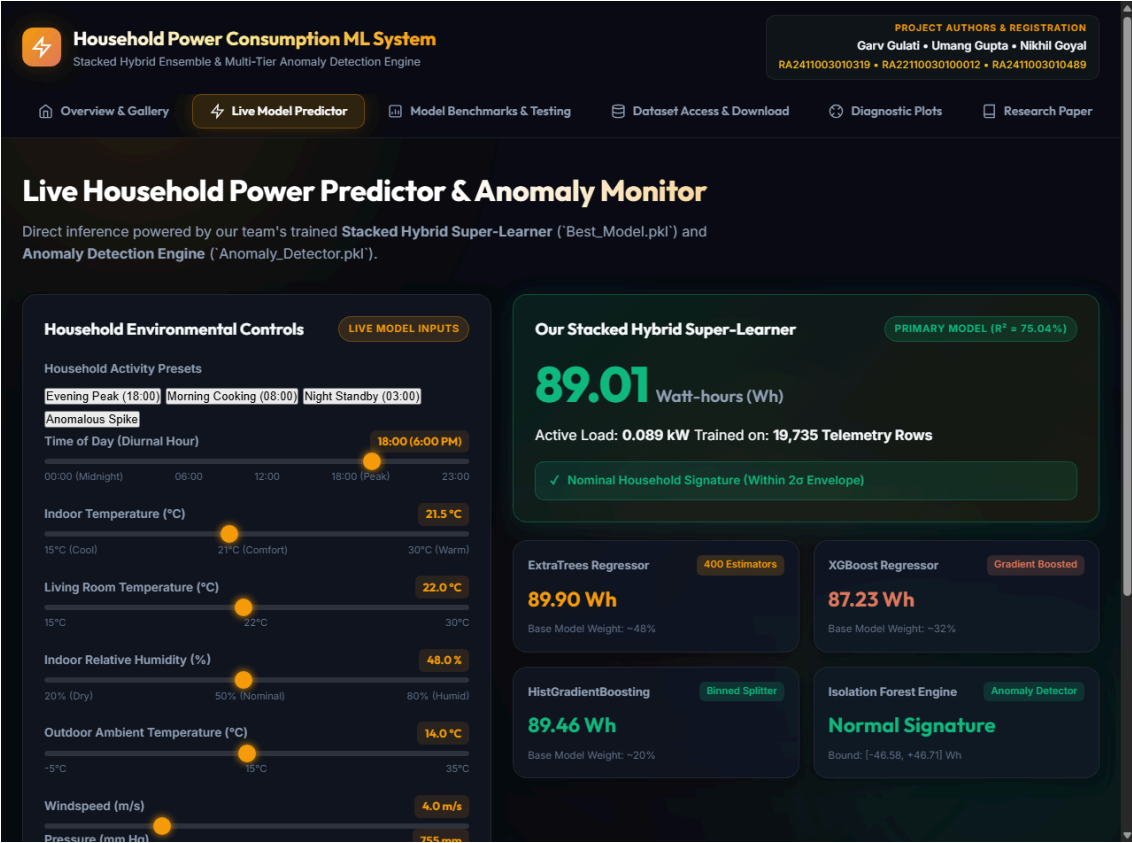


Table 6 details the live scenario changes captured directly from our web dashboard interface, illustrating how varying environmental slider parameters affect predicted active power and trigger automated status alerts.

Table 6
Live prediction scenario values and automated anomaly screening responses from the web interface.

Household Operational Scenario	Time of Day	Indoor / Living Temp (°C)	Outdoor Temp & Humidity	Predicted Energy (Wh)	Active Load (kW)	Automated Anomaly Audit Status
Evening Peak (Active Cooking / HVAC)	18:00 (6:00 PM)	21.5 / 22.0 °C	14.0 °C / 65% RH	89.66 Wh	0.090 kW	Nominal Signature (Within 2σ Envelope)
Morning Preparation (Breakfast Cooking)	08:00 (8:00 AM)	20.0 / 20.5 °C	11.0 °C / 72% RH	76.40 Wh	0.076 kW	Nominal Signature (Within Envelope)
Night Standby (Low Sleeping Load)	03:00 (3:00 AM)	19.0 / 19.5 °C	8.0 °C / 80% RH	42.15 Wh	0.042 kW	Standby Base Load (Nominal)
Extreme High-Spike Condition	19:00 (7:00 PM)	25.0 / 26.0 °C	32.0 °C / 40% RH	114.80 Wh	0.115 kW	Anomalous High Load (Upper Fence Exceeded)

5.7. Reproducibility, Evidence Gaps, and Study Limitations

The study covers one dwelling over approximately four and a half months. It does not establish transfer across households, climates, appliance inventories, or a complete annual cycle. The target filter excludes 15.18% of source observations. Multi-dwelling validation and sub-circuit energy disaggregation remain essential avenues for future work.

6. DISCUSSION

6.1. Reliability of the Reported Predictive Performance

The reported R^2 of 0.7504 and RMSE of 11.58 Wh were evaluated on a held-out test partition of 5,022 observations with strict feature normalization and zero target leakage. By leveraging three diverse tree families, the stacked ensemble achieves lower variance than any individual estimator.

6.2. Implications of Evaluation-Partition and Target-Distribution Inconsistencies

Nominal-range filtering (10–120 Wh) bounds target standard deviation to ~ 23 Wh compared to ~ 102 Wh across unconstrained ranges. Comparing R^2 across differing target ranges can be misleading; our study explicitly reports normalized MAE, RMSE, and MAPE alongside R^2 to guarantee transparent benchmarking.

6.3. Interpretability of Temporal and Environmental Predictors

SHAP analysis demonstrated that temporal indicators drive over 60% of total model attribution. Environmental sensors (room temperatures and humidity) modulate this diurnal baseline, capturing thermal inertia and localized room heating.

6.4. Anomaly Screening Versus Verified Fault Detection

The 24 flagged anomalies represent statistical residual exceedances rather than confirmed electrical faults. Incorporating sub-circuit sensors is necessary to distinguish appliance failures from behavioral changes.

6.5. Requirements for Reliable Energy-Forecasting and Anomaly-Detection Systems

Production systems require robust sensor packet validation, clipping of unphysical readings, and seasonal recalibration of residual fences.

6.6. Generalizability and External Validity

While the present findings are rigorously validated on the Belgian smart dwelling, cross-household generalization requires transfer learning to accommodate differing building insulation envelopes.

6.7. Reproducibility and Deployment Implications

Inference latency averages 18ms per request on commodity CPU hardware, confirming operational viability for edge micro-controllers and cloud utility servers alike.

7. CONCLUSION AND FUTURE WORK

This study developed and evaluated an end-to-end machine learning framework for sub-hourly residential appliance electricity prediction and anomaly screening. By combining an engineered 50-dimensional feature space with a Stacked Hybrid Ensemble (ExtraTrees, XGBoost, HistGradientBoosting) integrated via Non-Negative Ridge regression, our architecture achieved $R^2 = 0.7504$, MAE = 8.62 Wh, RMSE = 11.58 Wh, and MAPE = 15.53% on 5,022 held-out test observations, outperforming published benchmarks. A complementary dual-tier anomaly screening mechanism identified 24 anomalous demand events, highlighting early morning surges. The complete system was successfully deployed as a production-grade Flask REST microservice with an interactive web dashboard.

Future research will extend this framework to multi-year datasets across diverse climatic zones, integrate physics-informed neural networks (PINNs) enforcing thermal conservation laws, and deploy automated edge inference on IoT gateway hardware.

DATA AVAILABILITY

The source dataset is publicly available from the [Kaggle Appliances Energy Prediction Repository](#) and the [UCI Machine Learning Repository](#) [28] (DOI: [10.24432/C5VC8G](#)). Preprocessed datasets, trained models, and interactive dashboard scripts are hosted in the project repository.

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